

CM0832 - MT5001 Elements of Set Theory

Introduction

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2026-2

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

Pacto Pedagógico

Como miembros de la Universidad EAFIT, nos comprometemos a actuar de manera íntegra siguiendo los más altos estándares éticos y morales.

- Respeto
- Tolerancia
- Honradez
- Compromiso

Pacto Pedagógico

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-2/>

Pacto Pedagógico

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-2/>

Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

Pacto Pedagógico

Página web del curso

<https://asr.github.io/courses/cm0832-set-theory/2026-2/>

Conducto regular, fechas y porcentajes de las evaluaciones

La información está en la página web del curso.

Responsabilidad compartida

- Profesor
- Estudiantes

Asistencia a clase

Reglamento académico de los programas de posgrado, Título II, Capítulo VI, Artículo 62, Parágrafo 2.

“El estudiante de posgrado cuyas faltas de asistencia lleguen al treinta por ciento (30%) del total de las horas de clase programadas para el curso o para una parte de éste, cuando se desarrolle con más de un profesor, en secciones temáticas denominadas ‘módulo’, pierde con calificación de cero punto cero (0.0) del seminario o curso correspondiente y esta nota afecta el promedio crédito acumulado.”

Orientaciones para el curso

- Se recomienda cuatro horas de trabajo por semana (dos horas por cada hora de clase).
- Las clases son presenciales.
- La evaluación a la docencia es obligatoria.
- Se recomienda revisar periódicamente los canales de comunicación institucionales (EAFIT Interactiva y el correo institucional).
- El estudiante podrá entrar a clase a más tardar 20 minutos después de la hora asignada para su inicio.

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

Classic Set Theory

FOR GUIDED INDEPENDENT STUDY



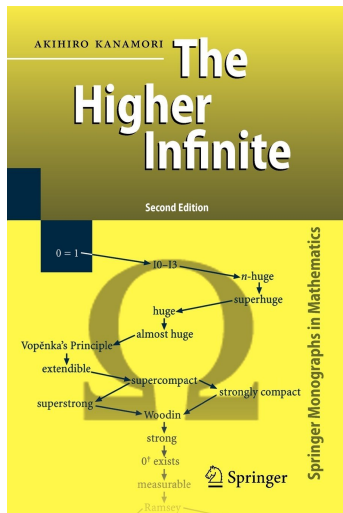
DEREK GOLDREI



CHAPMAN & HALL/CRC

Course Outline

- Introduction. Formal languages and theories. Methods of proof.
- The Zermelo-Fraenkel axioms.
- The axiom of choice.
- Ordered sets.
- Ordinal numbers.



*“Set theory **was born** on that December 1873 day when Cantor established that the reals are uncountable, i.e. there is no one-to-one correspondence between the reals and the natural numbers.” (Kanamori [1994] 2009, p. XII)*

Naive Set Theory

Remark

Cantor's set theory is also called **naive** set theory.

Naive Set Theory

Cantor's set definition

*“Unter einer **Menge** verstehen wir jede Zusammenfassung M von bestimmten wohlunterschiedenen Objecten m unsrer Anschauung oder unseres Denkens (welche die **Elemente** von M genannt werden) zu einem Ganzen.”*
(Cantor 1895, p. 481)

*“By an **aggregate** (Menge) we are to understand any collection into a whole M of definite and separate objects m of our intuition or our thought. These objects are called the **elements** of M .”* (Cantor [1895] 1915, p. 85)

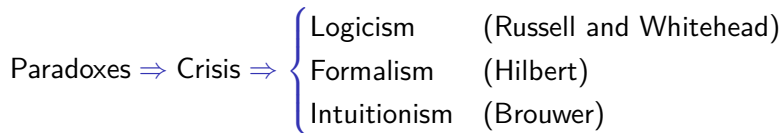
*“A **set** is a collection into a whole of definite, distinct objects of our intuition or our thought. The objects are called **elements (members)** of the set.”*
(Hrbacek and Jech [1978] 1999, p. 1)

Naive Set Theory

Transfinite numbers

Cantor introduced two kinds of **transfinite numbers**, each possessing its own arithmetic system that builds upon and expands the conventional arithmetic of natural number: **Ordinal** and **cardinal** numbers.

The Crisis in the Foundations of Mathematics



The Crisis in the Foundations of Mathematics

Logicism (Russell and Whitehead)

“The logicistic thesis is that mathematics is a branch of logic. The mathematical notions are to be defined in terms of the logical notions. The theorems of mathematics are to be proved as theorems of logic.” (Kleene [1952] 1974, p. 43)

The Crisis in the Foundations of Mathematics

Formalism (Hilbert)

“Classical mathematics shall be formulated as a formal axiomatic theory, and this theory shall be proved to be consistent, i.e. free from contradiction.”
(Kleene [1952] 1974, p. 53)

The Crisis in the Foundations of Mathematics

Intuitionism (Brouwer)

“Intuitionism is based on the idea that mathematics is a creation of the mind. The truth of a mathematical statement can only be conceived via a mental construction that proves it to be true.” (Iemhoff 2024)

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

First-Order Theories

Description

A **first-order theory** is a formal system built upon **first-order logic**, consisting of a set of **non-logic axioms** and the **theorems** that can be derived from them.

First-Order Theories

Description

A **first-order theory** is a formal system built upon **first-order logic**, consisting of a set of **non-logic axioms** and the **theorems** that can be derived from them.

First-order theories are characterised by their **non-logical symbols**, which includes constant symbols, function symbols and relation symbols.

First-Order Theories

Notation: Logical constants

$\wedge$	(and)	conjunction
$\vee$	(or)	(inclusive) disjunction
$\rightarrow$	(if __, then __)	conditional, material implication
$\neg$	(not)	negation
$\leftrightarrow$	(if and only if)	bi-conditional, material equivalence
$\forall x$	(for every x)	universal quantifier
$\exists x$	(there exists a x)	existential quantifier
$\exists! x$	(there exists one and only one x)	unique existential quantifier
$\doteq$	(equal)	identity, equality

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical symbols

The formal language $\mathcal{L}$ of the first-order theory of arithmetic (FA) is defined by

$$\mathcal{L} = \{', +, *, 0\}, \quad \text{where}$$

- (a) the symbol $'$ is an unary function symbol (successor function),
- (b) the symbol $+$ is a binary function symbol (addition function),
- (c) the symbol $*$ is a binary function symbol (multiplication function) and
- (d) the symbol 0 is a constant symbol (zero element).

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical symbols

The formal language $\mathcal{L}$ of the first-order theory of arithmetic (FA) is defined by

$$\mathcal{L} = \{', +, *, 0\}, \quad \text{where}$$

- (a) the symbol $'$ is an unary function symbol (successor function),
- (b) the symbol $+$ is a binary function symbol (addition function),
- (c) the symbol $*$ is a binary function symbol (multiplication function) and
- (d) the symbol 0 is a constant symbol (zero element).

That is, the symbols $\{', +, *, 0\}$ are **primitive (undefined)** concepts in FA.

(continued on next slide)

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical axioms of FA

$$\forall n \ 0 \neq n' \quad (\text{FA}_1)$$

$$\forall m \forall n (m' \doteq n' \rightarrow m \doteq n) \quad (\text{FA}_2)$$

$$\forall n \ n + 0 \doteq n \quad (\text{FA}_3)$$

$$\forall m \forall n (m + n' \doteq (m + n)') \quad (\text{FA}_4)$$

$$\forall n \ n * 0 \doteq 0 \quad (\text{FA}_5)$$

$$\forall m \forall n (m * n' \doteq (m * n) + m) \quad (\text{FA}_6)$$

For any unary property P ,

$$[P(0) \wedge \forall n (P(n) \rightarrow P(n'))] \rightarrow \forall n \ Pn \quad (\text{FA}_7) \text{ (axiom schema of induction)}$$

First-Order Theories

Example (first-order Dedekind-Peano arithmetic)

- Non-logical axioms of FA[†]

$$\forall n \ 0 \neq n' \quad (\text{FA}_1)$$

$$\forall m \forall n (m' \doteq n' \rightarrow m \doteq n) \quad (\text{FA}_2)$$

$$\forall n \ n + 0 \doteq n \quad (\text{FA}_3)$$

$$\forall m \forall n (m + n' \doteq (m + n)') \quad (\text{FA}_4)$$

$$\forall n \ n * 0 \doteq 0 \quad (\text{FA}_5)$$

$$\forall m \forall n (m * n' \doteq (m * n) + m) \quad (\text{FA}_6)$$

For any unary property P ,

$$[P(0) \wedge \forall n (P(n) \rightarrow P(n'))] \rightarrow \forall n \ Pn \quad (\text{FA}_7) \text{ (axiom schema of induction)}$$

[†]See, e.g. (Machover 1996; Hájek and Pudlák [1993] 1998; Skolem 1955; Robinson 1949).

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

Axiomatic Set Theories

Axiomatic set theories as a foundational system for mathematics

- *“Our axioms provide a sufficient collection of assumptions for the development of the whole of mathematics—a remarkable fact.” (Enderton 1977, p. 11)*
- *“But why axiomatize set theory in the first place? Well, for one thing, it is well known that set theory provides a unified framework for the whole of pure mathematics, and surely if anything deserves to be put on a sound basis it is such a foundational subject.” (Devlin [1979] 1993, p. 29)*
- *“Conventional mathematics is based on ZFC (the Zermelo-Fraenkel axioms, including the Axiom of Choice). Working withing ZFC, on develops... All the mathematics found in basic texts on analysis, topology, algebra, etc.” (Kunen [2011] 2013, p. 1)*

Axiomatic Set Theories

Some axiomatic systems of set theory

- Zermelo-Fraenkel set theory (ZF)
- Zermelo-Fraenkel set theory with Choice (ZFC)
- von Neumann-Bernays-Gödel set theory (NBG)
- Morse-Kelley set theory (MK)
- Tarski-Grothendieck set theory (TG)

But!

“On the other hand, we do not claim that every true fact about sets can be derived from the axioms we present. The axiomatic system is not complete in this sense.” (Hrbacek and Jech [1978] 1999, p. 3)

Foundations of Mathematics

Foundational systems

- (a) Set theories (with additional axioms)
- (b) Category theories
- (c) Type theories
- (d) Univalent foundations
- (e) Homotopy type theories

Foundations of Mathematics

Foundational systems[†]

- (a) Set theories (with additional axioms)
- (b) Category theories
- (c) Type theories
- (d) Univalent foundations
- (e) Homotopy type theories

[†]See, e.g. (Centrone, Kant and Sarikaya 2019).

Outline

Pácto Pedagógico

Introduction

First-Order Theories

Axiomatic Set Theories

References

References

- Georg Cantor (1895). Beiträge zur Begründung der transfiniten Mengenlehre. *Mathematische Annalen* 46.4, pp. 481–512. DOI: [10.1007/BF02124929](https://doi.org/10.1007/BF02124929) (cit. on pp. 15, 36).
- [1895] (1915). Contributions to the Founding of the Theory of Transfinite Numbers. Translation of Cantor (1895). Translated, and provided with an introduction and notes, by Philip E. B. Jourdain. Dover Publications (cit. on p. 15).
- Stefania Centrone, Deborah Kant and Deniz Sarikaya, eds. (2019). Reflections on the Foundations of Mathematics. General Thoughts. Vol. 407. Studies in Epistemology, Logic, Methodology, and Philosophy of Science. Springer (cit. on pp. 33, 34).
- Keith Devlin [1979] (1993). The Joy of Sets. Fundamentals of Contemporary Set Theory. 2nd ed. Undergraduate Texts in Mathematics. Springer-Verlag (cit. on p. 30).
- Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 30).
- Petr Hájek and Pavel Pudlák [1993] (1998). Metamathematics of First-Order Arithmetic. Second printing. Springer (cit. on pp. 27, 28).
- Karel Hrbacek and Thomas Jech [1978] (1999). Introduction to Set Theory. Third Edition, Revised and Expanded. Marcel Dekker (cit. on pp. 15, 32).

References

- Rosalie lemhoff (2024). Intuitionism in the Philosophy of Mathematics. In: The Stanford Encyclopedia of Philosophy. Ed. by Edward N. Zalta and Uri Nodelman. Summer 2024. Metaphysics Research Lab, Stanford University. URL: <https://plato.stanford.edu/archives/sum2024/entries/intuitionism> (cit. on p. 20).
- Akihiro Kanamori [1994] (2009). The Higher Infinite. Large Cardinals in Set Theory from Their Beginnings. 2nd ed. Springer (cit. on p. 13).
- Stephen Cole Kleene [1952] (1974). Introduction to Metamathematics. Seventh reprint. Bibliotheca Mathematica: A Series of Monographs on Pure and Applied Mathematics. Volume 1. North-Holland (cit. on pp. 18, 19).
- Kenneth Kunen [2011] (2013). Set Theory. Revised edition. Vol. 34. Mathematical Logic and Foundations. College Publications (cit. on p. 30).
- Moshé Machover (1996). Set Theory, Logic and their Limitations. Cambridge University Press (cit. on pp. 27, 28).
- Julia Robinson (1949). Definability and Decision Problems in Arithmetic. The Journal of Symbolic Logic 14.2, pp. 98–114. DOI: [10.2307/2266510](https://doi.org/10.2307/2266510) (cit. on pp. 27, 28).

References

Th. Skolem (1955). Peano's Axioms and Models of Arithmetic. In: Mathematical Interpretation of Formal Systems. Ed. by Th. Skolem et al. Vol. 16. Studies in Logic and the Foundations of Mathematics. North-Holland Publishing Company, pp. 1–14 (cit. on pp. 27, 28).